



EXTREMAL GRAPH THEORY

Erdős Problem 915

From Proof to Search

One-hour version

Louis Vandenbruwaene

Supervisor: Prof. Stijn Cambie

Promotor: Prof. Jan Goedgebeur

KU Leuven, Master of Mathematics

KU LEUVEN

The road

- 1 The question, and the one theorem it starts from
- 2 Three axes, twelve variants
- 3 What falls to a hand proof
- 4 The directed turn, where the hand proof stops
- 5 Why a machine, and what it is allowed to claim
- 6 Discovery: cooling and tabu
- 7 The landscape, the results, and what is left

Part 1

The question

How many edges can a graph on n vertices carry
if *no* pair of vertices has m edge-disjoint paths?

$$\ell_m(n) = \max \{ |E(G)| : |V(G)| = n, \lambda_G^{\max} \leq m - 1 \}$$

Mader, for simple graphs and $n \geq m$:

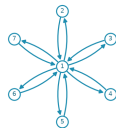
$$\ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor$$

Posed by Bollobás and Erdős. Menger turns “many routes” into “costly cut”, and the whole thesis is built on that one exchange.

What an extremal graph looks like

Extremal graphs of the named families, drawn large

multigraph directed, arc
 $m = 3, n = 7$: double star (24 arcs)



all arcs $\times 2$

simple directed, arc
 $m = 2, n = 7$: bidirected star (12 arcs)



multigraph directed, arc
 $m = 3, n = 7$: bipartite wall $2 \cdot B_{1,1}$ (24 arcs)



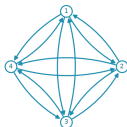
all arcs $\times 2$

multigraph undirected, edge
 $m = 3, n = 9$: star at multiplicity 2 (16 edges)



all edges $\times 2$

simple directed, arc
 $m = 4, n = 4$: bidirected K_4 (12 arcs)



hypergraph, $r = 4$
 $m = 2, n = 13$: star hypertree (4 hyperedges)



hypergraph, $r = 3$
 $m = 2, n = 16$: star hypertree (7 hyperedges)



hypergraph, $r = 3$
 $m = 2, n = 10$: star hypertree (4 hyperedges)



directed hypergraph, $r = 3$
 $m = 2, n = 7$: tail-to-heads star



The named families, each sitting exactly on its variant's ceiling.

Part 2

Three axes, twelve variants

The three axes

Model

simple graph
multigraph
 r -uniform hypergraph

arity and multiplicity, two independent toggles

Direction

undirected
directed

direction breaks the degree symmetry
that caps the undirected answer

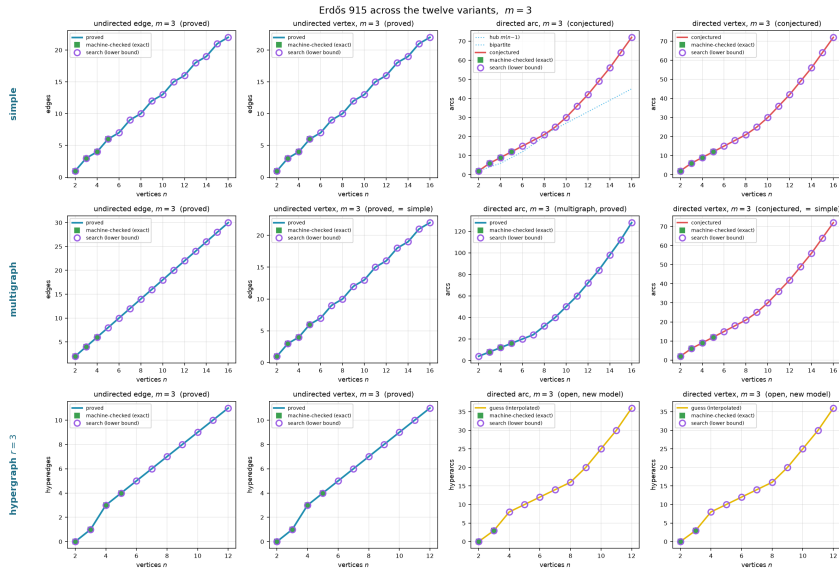
Separation

edge-disjoint, $\lambda^{\max}$
internally vertex-disjoint, $\kappa^{\max}$

Whitney: $\kappa \leq \lambda$, so the
vertex-feasible family is the *larger* one

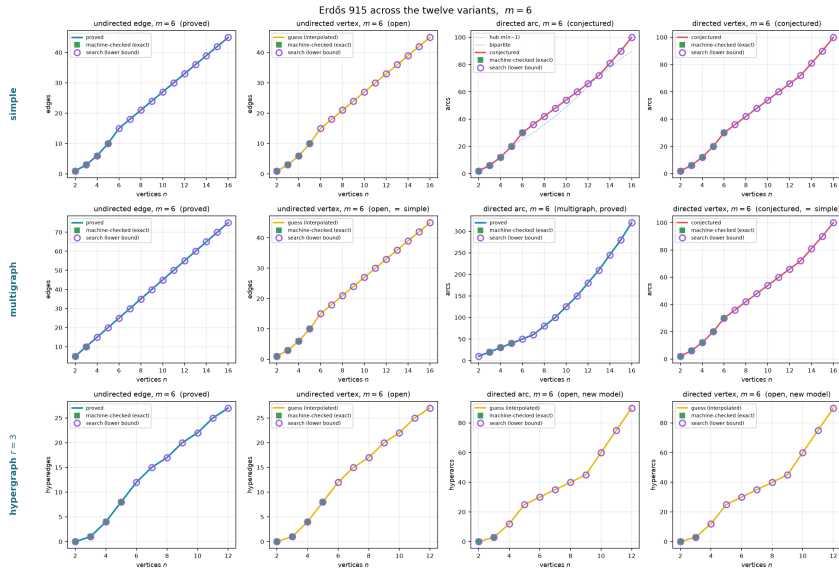
$3 \times 2 \times 2 = 12$ versions of one question

Twelve variants at $m = 3$



Blue proved for all n , red conjectured, yellow interpolated. Squares proved exactly, circles are lower bounds.

The same twelve at $m = 6$



Raising the threshold moves the curves, not the regime. The undirected vertex panels have now gone open.

Part 3

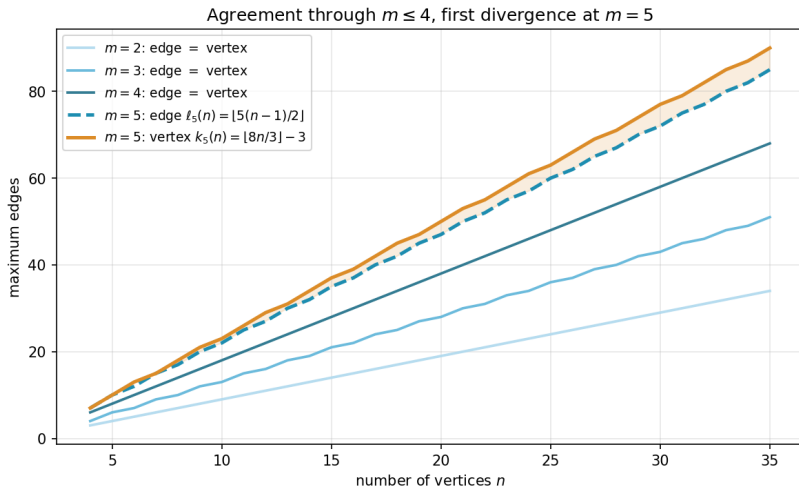
What falls to a hand proof

The short proofs

Multigraphs, edge-disjoint	$L_m(n) = (m - 1)(n - 1)$
Simple graphs, vertex-disjoint at $m = 2$	$k_2(n) = n - 1$, exactly the trees
Simple graphs, $m \leq 4$, $n \geq m$	$k_m(n) = \ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor$ (Leonard)
Hypergraphs, edge-disjoint	$\ell_m^{(r)}(n) = \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$
Random graphs	threshold $p^* = m/n$

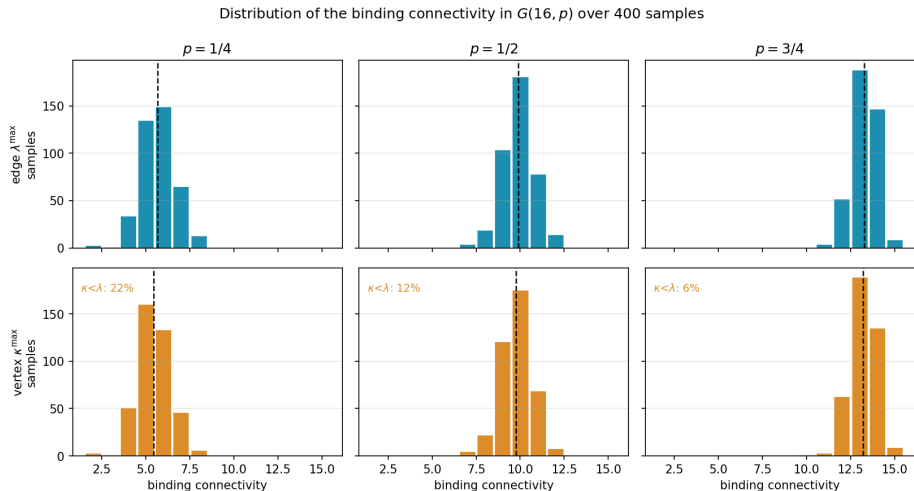
One argument serves the first and the fourth: charge every edge to the Gomory–Hu tree, a summary of all $\binom{n}{2}$ connectivities in $n - 1$ numbers.

The first divergence, at $m = 5$



Edge and vertex agree through $m = 4$. At $m = 5$ they part, and beyond it the vertex problem has stood open since 1974.

The same split, visible in random graphs



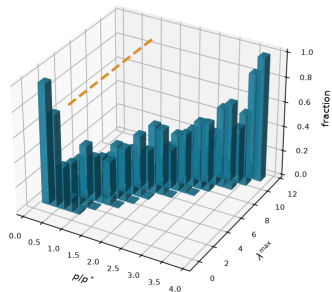
$G(16, p)$ at three densities. The vertex distribution never sits right of the edge one, which is Whitney across a whole population, and the gap closes as the graph fills in.

Where the forbidden pair appears

Distribution of $\lambda^{\max}$ against density, $m = 3$, each model in units of its own threshold p^* (the density at which a vertex's expected degree reaches m)

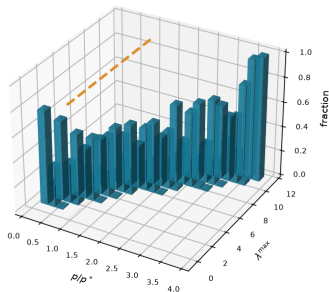
undirected edge ($n = 12$)

$$p^* = m/n = 3/12$$



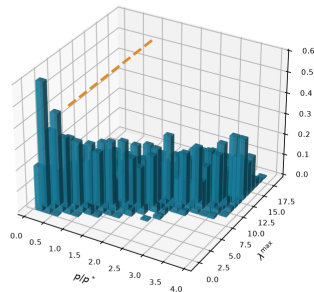
directed arc ($n = 12$)

$$p^* = m/n = 3/12$$



hypergraph edge ($r = 3, n = 8$)

$$p^* = m/\binom{r}{2} = 3/21$$

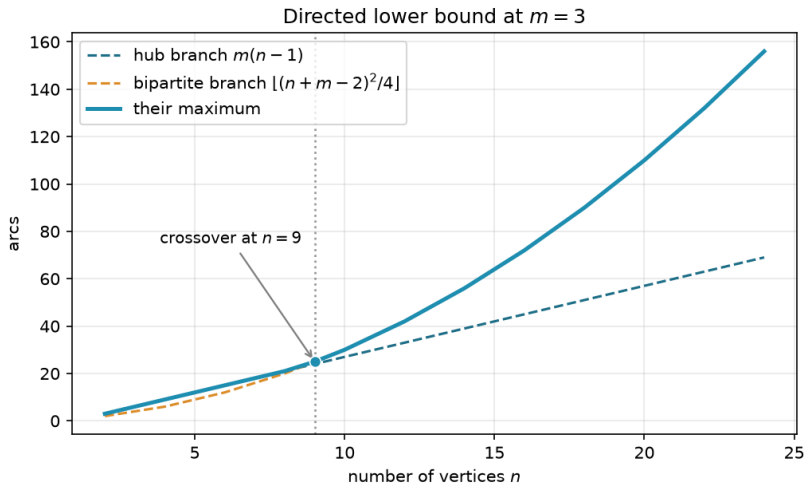


Each model swept in units of its *own* threshold, since a hypergraph vertex sits in $\binom{n-1}{r-1}$ hyperedges and so lives on a different density scale entirely.

Part 4

The directed turn

Linear king against quadratic cabinet



The hub spends its arcs on one centre and grows linearly. The one-directional bipartite wall grows quadratically and takes the lead at $n = 9$ when $m = 3$.

The directed conjecture, and what is unconditional

Conjectured value, lower bound proved for all m :

$$\ell_m^{\text{dir}}(n) = \max\left(m(n-1), \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor\right)$$

Proved here with no hypothesis on the extremal shape:

$$|A(D)| \leq \left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)(n-1)$$

and therefore

$$\ell_m^{\text{dir}}(n) = \frac{n^2}{4} + \Theta_m(n)$$

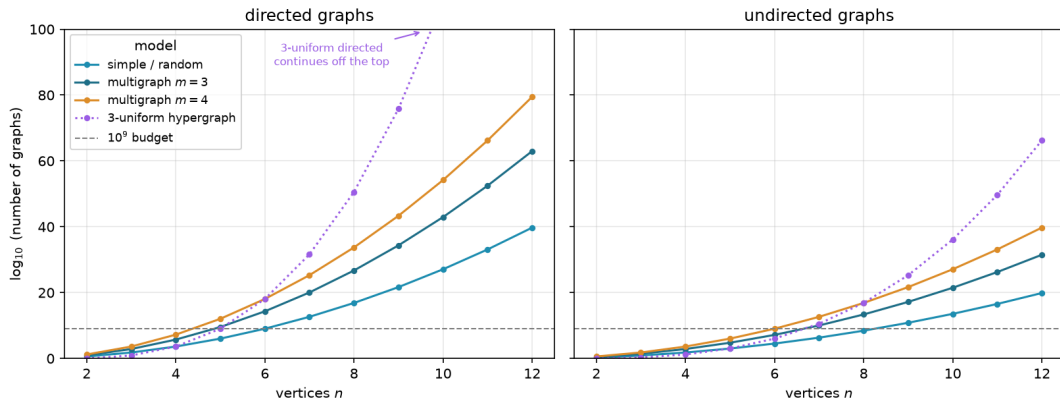
Leading constant and order of the second term: both now theorems. What is left is one coefficient, $(m-2)/2$ conjectured against $4(m-1)$ proved.

Part 5

Why a machine

Enumeration detonates

Number of graphs blind enumeration must visit



A simple graph offers 2^E candidates, a multigraph m^E . Every curve passes a generous budget while n is in single digits, and the directed cases go first.

Three jobs, kept apart

MEASURE

One integer maximum flow, on one multiplicity matrix, for all twelve variants.

What it reports about a given graph is a fact.

PROVE

Exhaust every graph of a fixed size, or run the cut-counting program.

A finished sweep is an upper bound with the standing of a hand proof.

DISCOVER

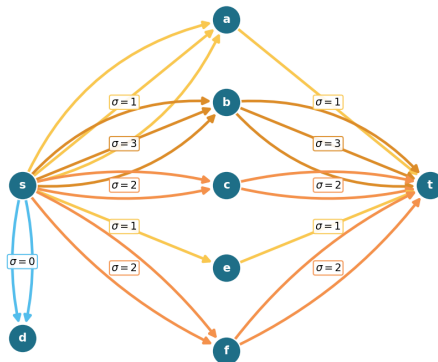
Cool a random graph, or climb greedily with a tabu list.

Only ever exhibits a graph. Lower bounds only. Never a proof.

The honesty contract: the search proposes a lower bound, a proof or a finished exhaustion disposes the upper bound, and only when the two meet is a value called settled.

The dial the search turns

Parallel arcs drawn explicitly, coloured by sensitivity σ :
cool ($\sigma = 0$, free) $\rightarrow$ warm (load-bearing)



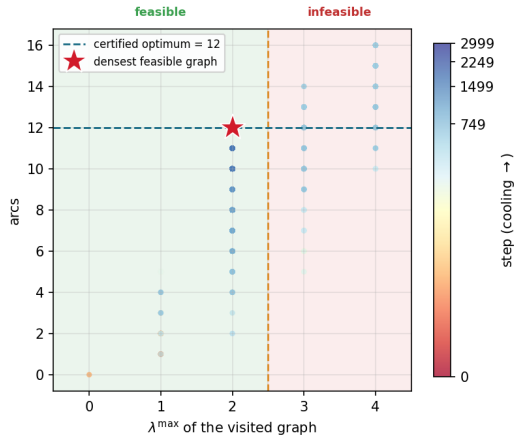
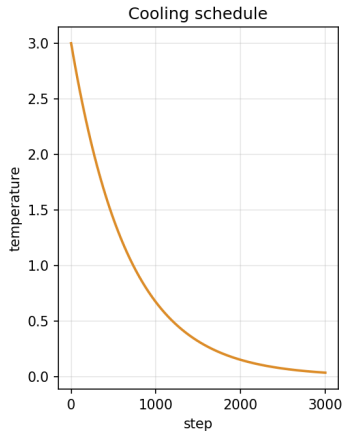
Edge sensitivity $\sigma(e) = \lambda^{\max}(G) - \lambda^{\max}(G - e)$. Both $s \rightarrow a$ and $s \rightarrow b$ carry three parallel arcs, yet only b is load-bearing. Multiplicity is not the same as carrying weight.

Part 6

Discovery

One cooling run

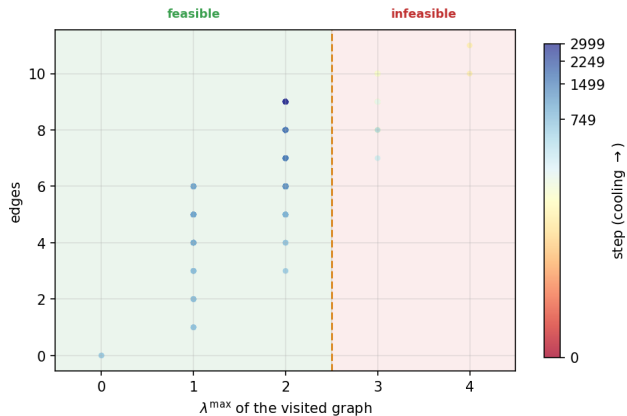
Cooling toward an extremal graph



Every visited graph by connectivity and size, coloured by step. Hot it strays above the ceiling and picks up arcs it may not keep, cool it is driven back onto the proved optimum.

The same behaviour, undirected and simple

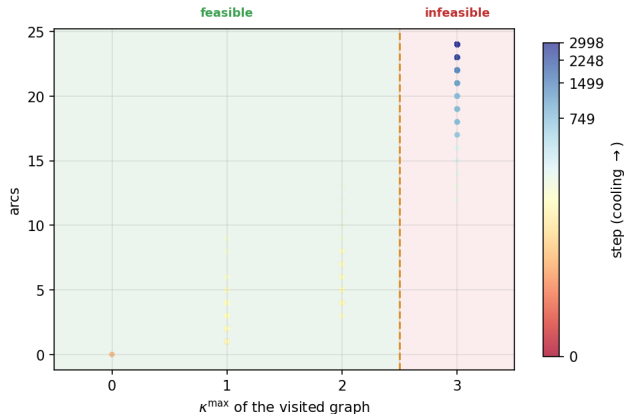
Search trace: simple undirected, $n = 7$, $m = 3$



Simple undirected, $n = 7$, $m = 3$.

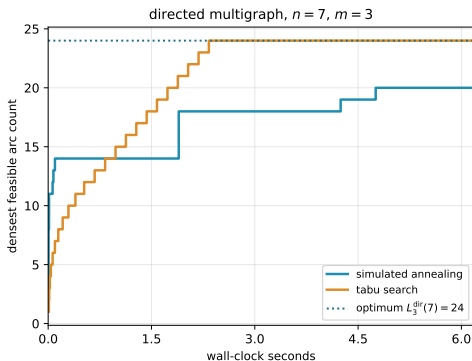
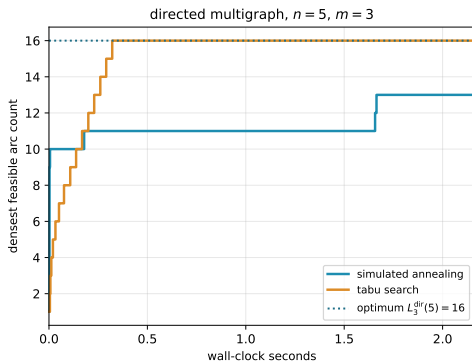
And under the stricter separation

Search trace: multi directed vertex-sep, $n = 4$, $m = 3$



Directed multigraph, $n = 4$, $m = 3$, vertex-disjoint. The walk lingers far longer on the infeasible side, which is what a harder constraint looks like from inside the search.

Cooling against tabu



Variant	n	m	target	SA in 6 s	tabu in 6 s
simple undirected	7	3	9	9	9
simple directed	7	3	18	16	18
directed multigraph	7	3	24	20	24

Tabu takes fewer, better-aimed steps and assembles the structured extremiser the cooled walk cannot reach in budget. Only 9 is proved (Mader); 18 and 24 are best known, not yet proved.

Validation by rediscovery

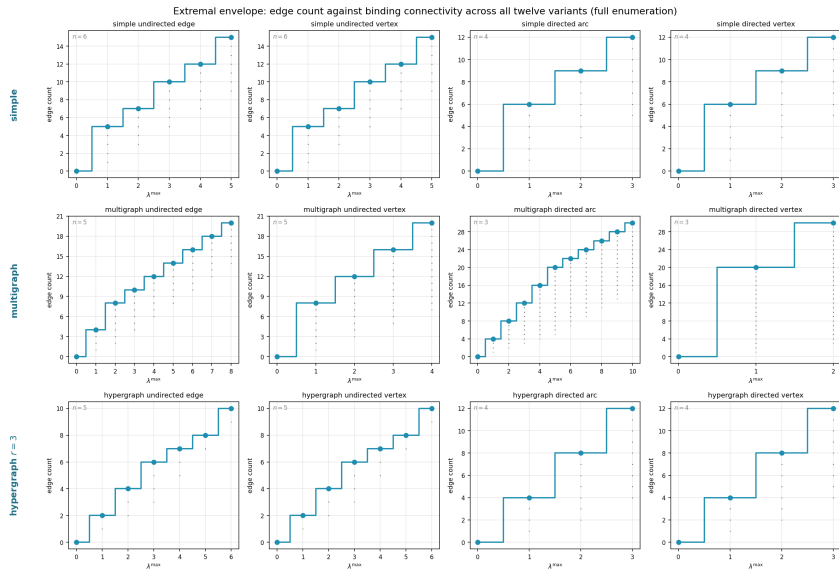
Construction	n	m	Search	Theory
spanning tree (simple undirected, $m=2$)	6	2	5	5
multiplicity- $(m-1)$ star (multi undirected, $m=3$)	6	3	10	10
bidirected star (simple directed, $m=2$)	4	2	6	6
bidirected star (simple directed, $m=2$)	6	2	10	10
hub-bipartite tie (simple directed, $m=2$)	7	2	12	12
double star (multi directed, $m=3$)	4	3	12	12
double star (multi directed, $m=3$)	5	3	16	16

Pointed at the cases already settled and told only to pack in edges, the search returns the named constructions: the spanning tree, the thickened star, the double star, the bidirected star, and the hub-bipartite tie at $n = 7$.

Part 7

The landscape, and what is left

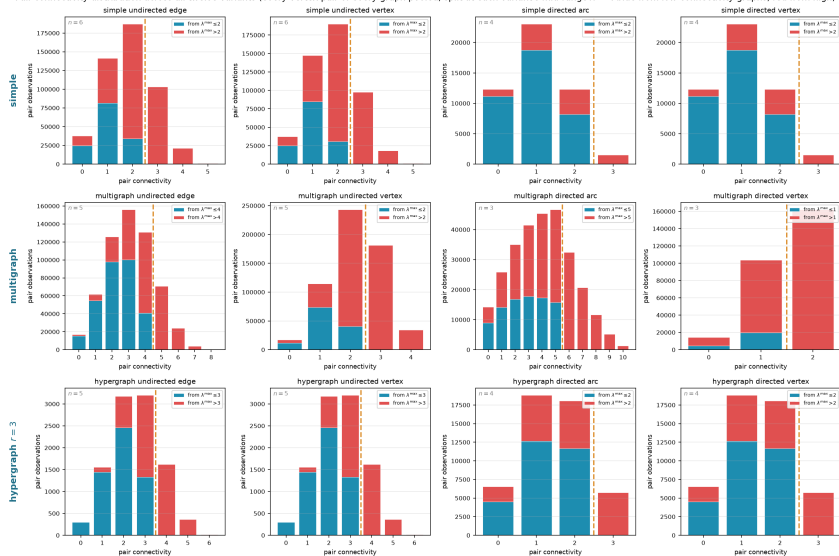
Every graph at small n , and the frontier



One point per graph. The blue staircase is the extremal envelope, and the theorems forbid everything above it.

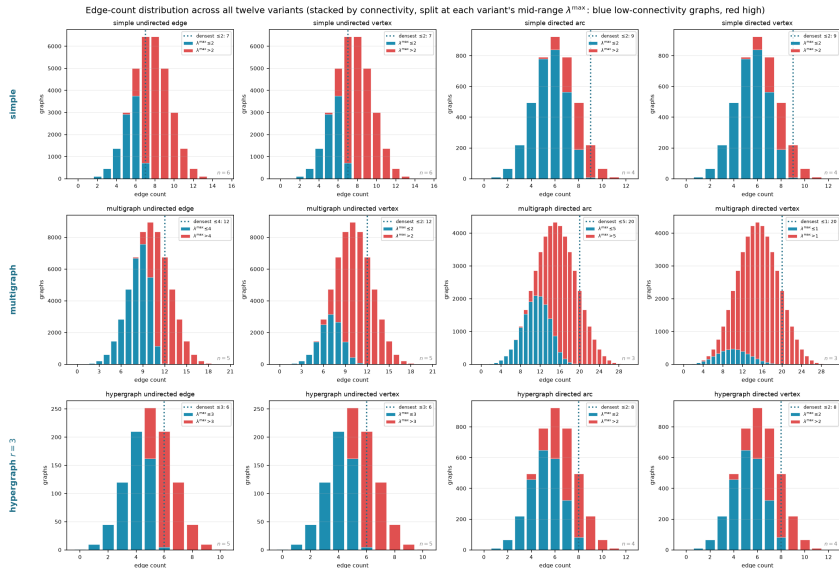
Inside each graph, pair by pair

Pair-connectivity distribution across all twelve variants (every vertex pair of every graph pooled, split at each variant's mid-range λ^{max} ; blue from low-connectivity graphs, red from high)



High-connectivity pairs are rare even inside the graphs that have one.

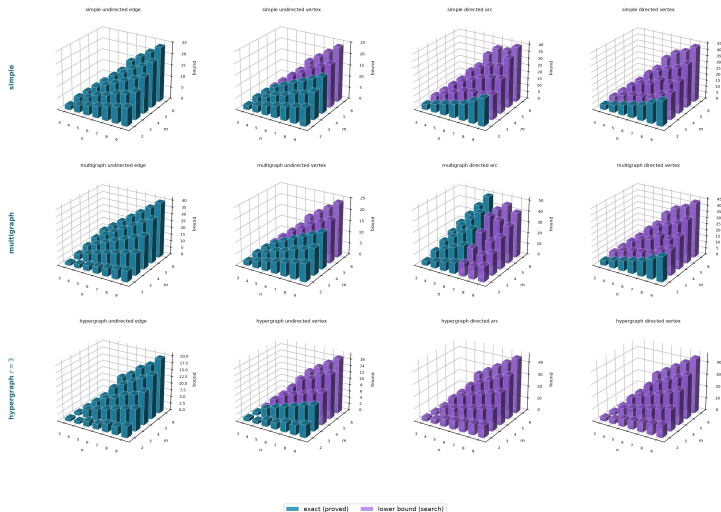
The same graphs, by sheer size



The dotted line marks the densest low-connectivity graph. It sits far out in the sparse tail.

The whole study as one surface

Erdős 915: optimal bound over (n, m) , blue where proved, purple where only a search lower bound is known



Blue is machine-proved, purple is a search lower bound.

Where each variant stands

Separation	Model	Value	Status
Undirected			
edge	simple	$\left\lfloor \frac{m(n-1)}{2} \right\rfloor$	PROVED Mader
edge	multi	$(m-1)(n-1)$	PROVED
edge	hyper	$\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$	PROVED Gomory–Hu
vertex	simple	$\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ ($m \leq 4$), $\lfloor 8n/3 \rfloor - 3$ ($m = 5$, $n \geq 13$)	CITED open $m \geq 6$
vertex	multi	adjacency count, so the simple value	convention
vertex	hyper	$\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ ($m \leq 3$)	PROVED $m \leq 3$, open $m \geq 4$
Directed			
arc	simple	$\max(2(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor)$ ($m = 2$), $n^2/4 + \Theta_m(n)$ (all m)	PROVED $m = 2$, CONJ $m \geq 3$
arc	multi	$(m-1) \max(2(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor)$	PROVED every n and m
arc	hyper	$(1 + o(1))(m-1)n^2/(4(r-1))$	LEADING TERM exact value open
vertex	simple	$\max(2(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor)$ ($m = 2$), $n^2/4 + \Theta_m(n)$ (all m)	PROVED $m = 2$ separately, CONJ $m \geq 3$
vertex	multi	adjacency count, so the simple digraph value	convention
vertex	hyper	$(1 + o(1))(m-1)n^2/(4(r-1))$	LEADING TERM exact value open

PROVED holds for every n . **CONJ** conjectured formula, with a matching lower bound proved and the small cases exhausted.

LEADING TERM the n^2 term is a theorem, the exact value is not.

What is new here

1 The hypergraph vertex problem, settled at $m = 2$ and $m = 3$

for every uniformity r and every n : $k_m^{(r)}(n) = \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor = \ell_m^{(r)}(n)$. The $m = 3$ case rests on a new rank bound for incidence graphs, proved through Tutte's triconnected decomposition.

2 One-way simple graphs: $\ell_m^{\text{dir}}(n) = k_m^{\text{dir}}(n) = n^2/4 + \Theta_m(n)$, unconditionally

no assumption about the extremal shape, which every earlier route to this bound needed. The vertex separation is not a corollary of the arc one, and gets the same bound by borrowing the route family rather than the conclusion. Leaves a constant factor of about eight in one coefficient.

3 One-way multigraphs, solved outright:

$$L_m^{\text{dir}}(n) = (m-1) \max\left(2(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor\right)$$

every n and every m , by peeling off one sparse reachability-preserving subgraph per unit of $m-1$. No parity split, no auxiliary conjecture, and the finite $n = 7$ fact this problem once needed a solver for now follows in three lines.

4 One-way hyperedge networks: the leading term settled

$(1 + o(1))(m-1)n^2/(4(r-1))$, for edge- and vertex-disjoint routes alike, so the bipartite construction is asymptotically optimal.

What is left

- **The undirected vertex problem at $m \geq 6$** , open since 1974, extremal structure genuinely unknown
- **The missing decomposition**: that an extremal non-hub digraph splits as $A \cup B$ with every arc from A to B , none inside A , none coming back. Four hypotheses, of which the backward arc is the hardest.
- **The directed vertex problem at $m \geq 3$** , which Whitney's inequality pointedly does *not* hand over with the arc case. Its leading term is settled all the same, by borrowing the route family rather than the bound
- **The exact values** behind the settled leading terms: the linear coefficient one-way, and the hyperedge value at finite n

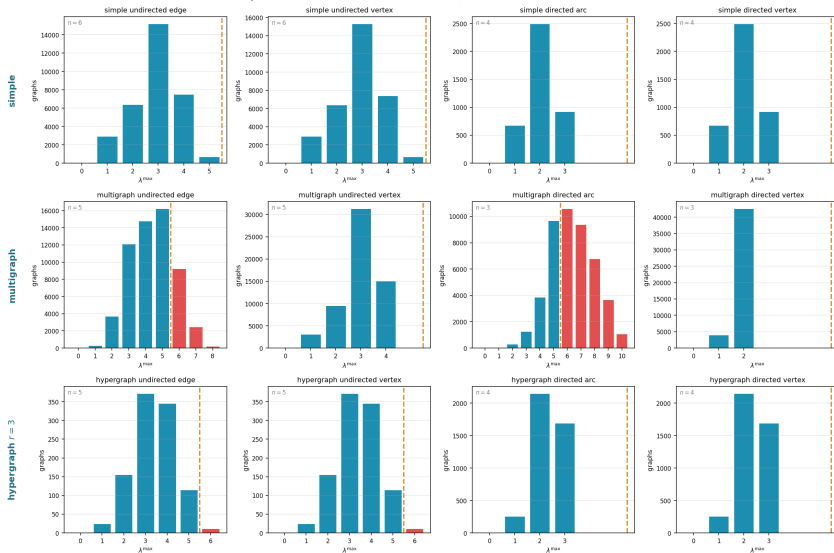
The map is filled in where honest mathematics could fill it,
and where it could not, the blanks are marked precisely.

Backup

Further figures for the question round

The distribution at a higher threshold

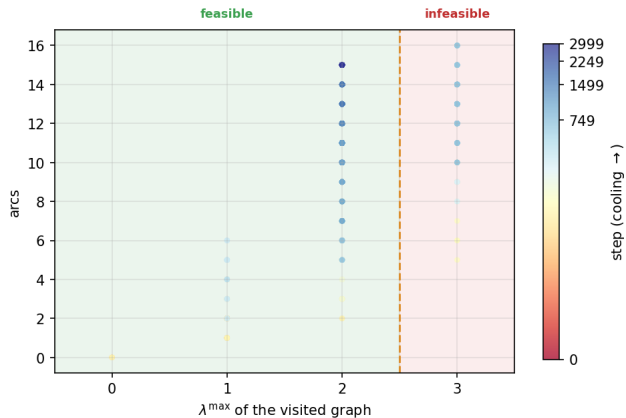
Connectivity distribution across all twelve variants, $m = 6$ (blue = feasible $\lambda^{\max} \leq 5$, red = infeasible)



At $m = 6$ the feasibility boundary sits out past the bulk of the distribution.

Trace: directed multigraph

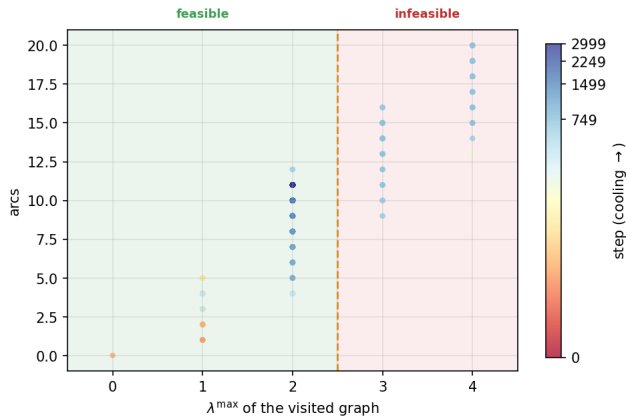
Search trace: multi directed, $n = 5$, $m = 3$



Directed multigraph, $n = 5$, $m = 3$.

Trace: simple directed

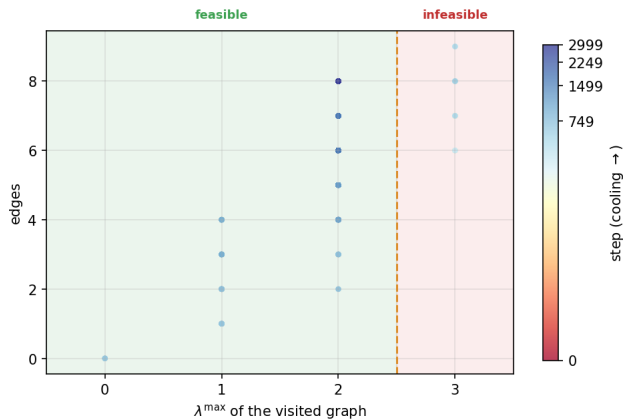
Search trace: simple directed, $n = 5$, $m = 3$



Simple directed, $n = 5$, $m = 3$.

Trace: multigraph undirected

Search trace: multi undirected, $n = 5$, $m = 3$



Multigraph undirected, $n = 5$, $m = 3$.

Orientation models for the directed hypergraph

r	m	n	forward	general
3	3	3	3	4
3	3	4	8	8
4	3	4	4	6
4	4	4	4	8

Backward equals forward by arc reversal, and the general model obeys the same cap. The general model is strictly denser only while vertices are scarce, $n \approx r$. Once n exceeds r the computed gap closes, so the orientation axis appears to collapse (computational evidence, not proved) and Problem 915 stays a twelve-variant question.

The multigraph vertex problem, other convention

		$n = 2$	3	4	5	6	7
$m = 2$	value	1	2	3	4	5	6
	tree	1	2	3	4	5	6
$m = 3$	value	2	4	6	8	10	12
	tree	2	4	6	8	10	12
$m = 4$	value		6	9	12	15	
	tree		6	9	12	15	
	theta		6	9	12		
$m = 5$	value			14	19		
	tree			12	16		
	theta			14	19		
$m = 6$	value			19			
	tree			15			
	theta			19			

Reading parallel copies as distinct routes makes this a genuinely new problem. Bold entries beat the multigraph edge value. The natural guess, that a thickened tree wins once $n \geq m + 2$, is false for every $m \geq 5$: a bouquet of thickened cliques sharing one vertex beats it by a margin growing linearly in n .